Code No: 136AW **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
CRYPTOGRAPHY AND NETWORK SECURITY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Discuss the need of security.	[2]
b)	What is steganography?	[3]
c)	Write about cipher block chaining mode of operation.	[2]
d)	List out the advantages of RC5 algorithm.	[3]
e)	Define CMAC.	[2]
f)	Explain Kerberos.	[3]
g)	Discuss the functions of SSH protocol.	[2]
h)	Discuss about the importance of security in mobile devices.	[3]
i)	What is transport mode and tunnel mode in IP sec?	[2]
j)	What is authentication header?	[3]

PART – B

(50 Marks)

2.a)	List and briefly define categories of Security Services and attacks.	
b)	Discuss about Brute force attack in detail.	[5+5]
	OR	
3.a)	Demonstrate the terminologies used in Encryption.	
b)	Discuss about model for Network Security.	[5+5]
4.a)	Critically analyze the security of RSA.	
b)	Enumerate in detail about the steps in Blow Fish Algorithm and explain the process of each round with a neat diagram.	[5+5]
	OR	
5.a)	Differentiate between RC5 and blowfish.	
b)	Write a short note on knapsack algorithm.	[5+5]
6.a)	Discuss about Secure Hash algorithm.	
b)	Explain about Digital Signature Scheme.	[5+5]
	OR	
7.a)	Discuss about the message exchange mechanism in Kerberos version 4.	
b)	Explain about X.509 authentication service.	[5+5]

8.a) List and briefly define the parameters that define an SSL session state.
b) Discuss briefly about HTTPS. [5+5]

OR

9.a) Discuss the IEEE 802.11i Wireless LAN Security.
b) What are web security consideration? Explain. [5+5]

10.a) Describe the functionalities of Internet Key Exchange Protocol.
b) What is a firewall? State the design goals of firewalls. Also discuss in detail different types of firewalls. [5+5]

OR

11.a) How inter branch payment transactions are performed?
b) Explain in detail about the mechanism of security associations and IP Security in IPSec. [5+5]

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